

Policy relevant challenges for flexibility, reliability and cost of advanced biofuels and their risk (low-green, moderate-yellow, high-red) towards market uptake for 2030.

	Flexibility	Reliability	Cost
Biomass supply			
Agriculture & forest residues	Soil carbon loss	Biodiversity loss	Disperse, low density, ununiform material [152]
Lignocellulosic crops	Competing markets Spread of invasive species Degraded land improvement	Nitrogen leaching due to overharvest Direct & indirect land use change	Competing markets Low yield (high production cost per unit) in degraded land
Organic wastes & lipids	Limited bulk availability	Variable quality	Collection and sorting
Conversion pathways			
Fermentation	Co-location with existing infrastructure	Establishing adequate operational capacity by 2030	Establishing adequate operational capacity by 2030 Process efficiency, especially for butanol production By-products utilisation (e.g., lignin fraction)
Gasification		Gas conditioning and clean up Co-location	
Pyrolysis	Pyrolysis oil upgrading and/or coprocessing	Capacity redundancy required for First-of-A-Kind plants Catalysts with improved selectivity and stability for pyrolysis oil upgrading are not yet commercial	Uncertain production costs Pyrolysis oil upgrading and/or coprocessing
Upgrading and hydrotreatment	Co-location with existing infrastructure	Establishing adequate operational capacity by 2030	Establishing adequate operational capacity by 2030 Process efficiency, especially for butanol production By-products utilisation (e.g., lignin fraction)
End Use			
Aviation	SAF must be drop-in	Competition with biodiesel production	International competition with non-EU operators
	Blending restricted to 50% (safety and lubricity restrictions) Lower aromatics content in some of SAFs might lead to jet engine compatibility issues (older engines)	Fuel quality critical to ensure safety in the air – variations in fuel quality dependent on conversion and feedstock	Fossil kerosene cheaper than all certified SAF pathways Upgrading of bio-oil/biocrude leads to increased final price
Marine	Competition with biodiesel production Existing fuel infrastructure, no engine modifications	International competition with non-EU operators Access to renewable raw materials	International competition with non-EU operators
	Ensuring compatibility of new fuels or their blends & blending behaviour Dedicated infrastructure needed onboard as well as in harbour (in case of methanol or LBG)	Fuel quality of less refined fuel should enable year-round operation High fuel volumes needed for single vessel – risk of advanced biofuel seasonal shortage	Low prices of HFO, MGO
Road	Multi fuel blends increase complexity Use of existing fuel infrastructure, no engine modifications	Access to renewable raw materials	Loss of cargo space due to bigger fuel tanks
	Infrastructure and engine modifications needed for higher alcohols or FAME blends. Completely new infrastructure for DME or hydrogen	Shortage of sustainable feedstock for HVO or ethanol production	International competition with non-EU operators Need of compensation for lower income end-users Price gap between advanced biofuels and fossils End-user choosing economically justified option

impacts to direct and indirect land use change [186].

way forward for sustainable, long term biomass supply practices to